An empirical example: Four terminals and two areas

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2024 / 07 / 29

```r
# Four taxa and two areas


## Libraries

library(blepd)
```

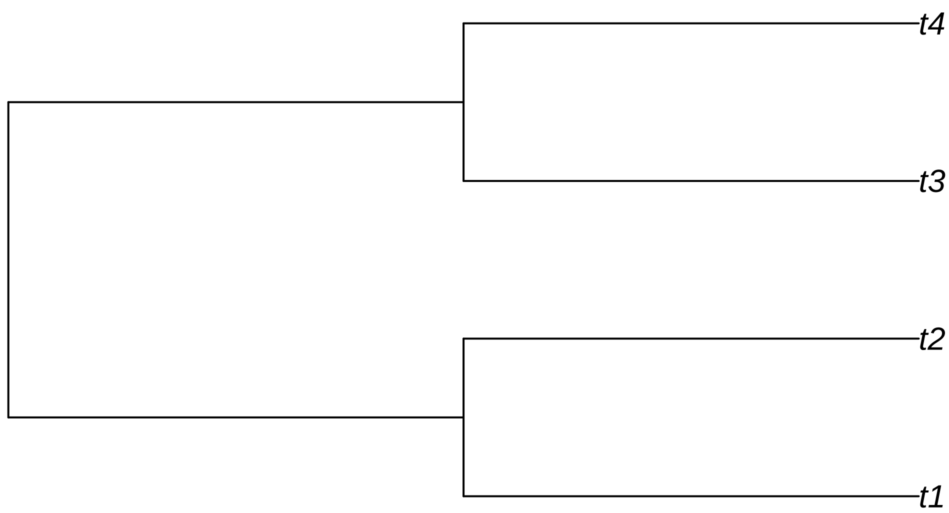
```
## Loading required package: ape
## Loading required package: picante
## Loading required package: vegan
## Loading required package: permute
##
## Attaching package: 'permute'
## The following object is masked from 'package:devtools':
##
##     check
## Loading required package: lattice
## This is vegan 2.6-6.1
## Loading required package: nlme
```

```r
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
##
## If you use the ggtree package suite in published research, please cite
## the appropriate paper(s):
##
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
## ggtree: an R package for visualization and annotation of phylogenetic
## trees with their covariates and other associated data. Methods in
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
##
## Guangchuang Yu, Tommy Tsan-Yuk Lam, Huachen Zhu, Yi Guan. Two methods
## for mapping and visualizing associated data on phylogeny using ggtree.
## Molecular Biology and Evolution. 2018, 35(12):3041-3043.
## doi:10.1093/molbev/msy194
##
## Shuangbin Xu, Lin Li, Xiao Luo, Meijun Chen, Wenli Tang, Li Zhan, Zehan
## Dai, Tommy T. Lam, Yi Guan, Guangchuang Yu. Ggtree: A serialized data
```

```
## object for visualization of a phylogenetic tree and annotation data.
## iMeta 2022, 1(4):e56. doi:10.1002/imt2.56
##
## Attaching package: 'ggtree'
## The following object is masked from 'package:nlme':
##
##     collapse
## The following object is masked from 'package:ape':
##
##     rotate
```

```r
library(ggplot2)



## Data

# Call the 4 terminals example

##data(package = "blepd")

## trees

data(tree)

#str(tree)


plot(tree,use.edge.length =T)
```

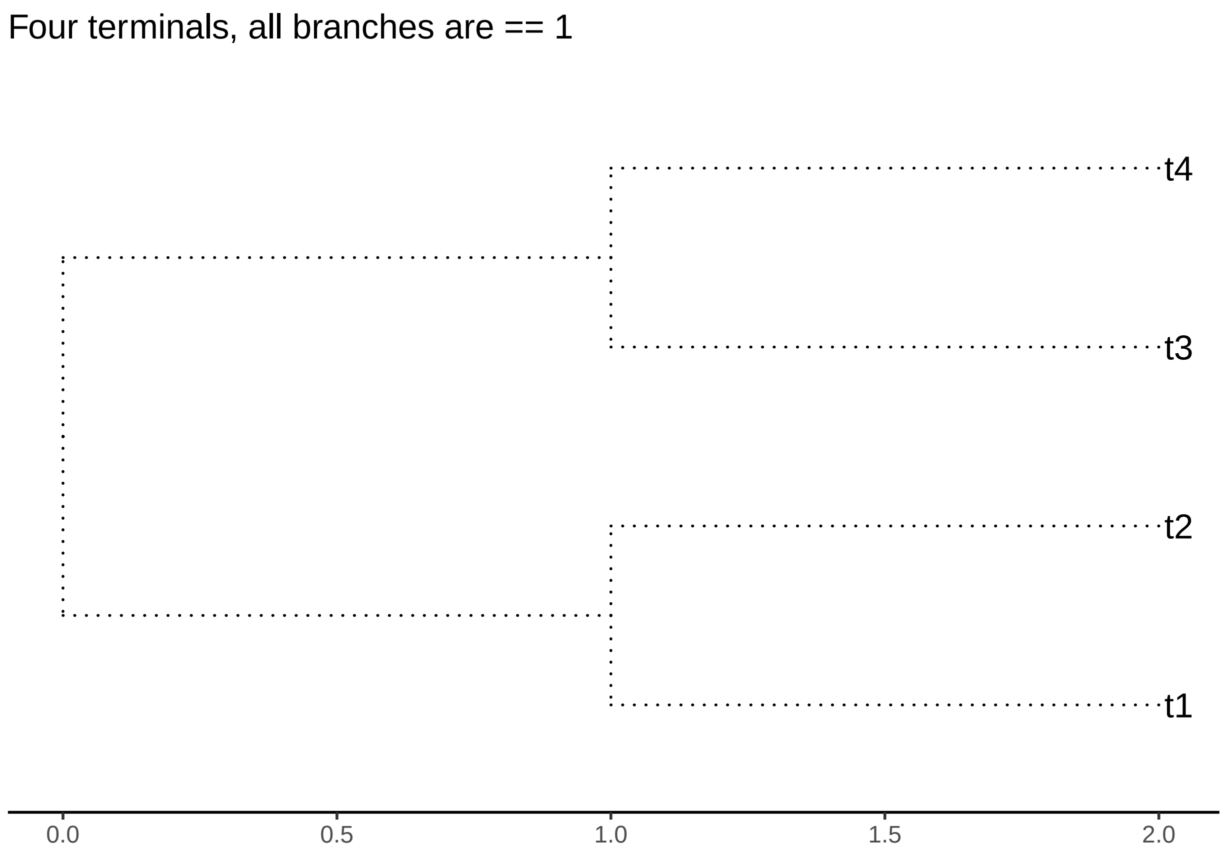



```r
initialTree <- tree


## distributions

data(distribution)

##str(distribution)

dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY
```

```
##   Terminal Area
## 1       t1   A1
## 2       t3   A1
## 3       t2   A2
## 4       t4   A2
```

```r
## plotting

## the tree
```

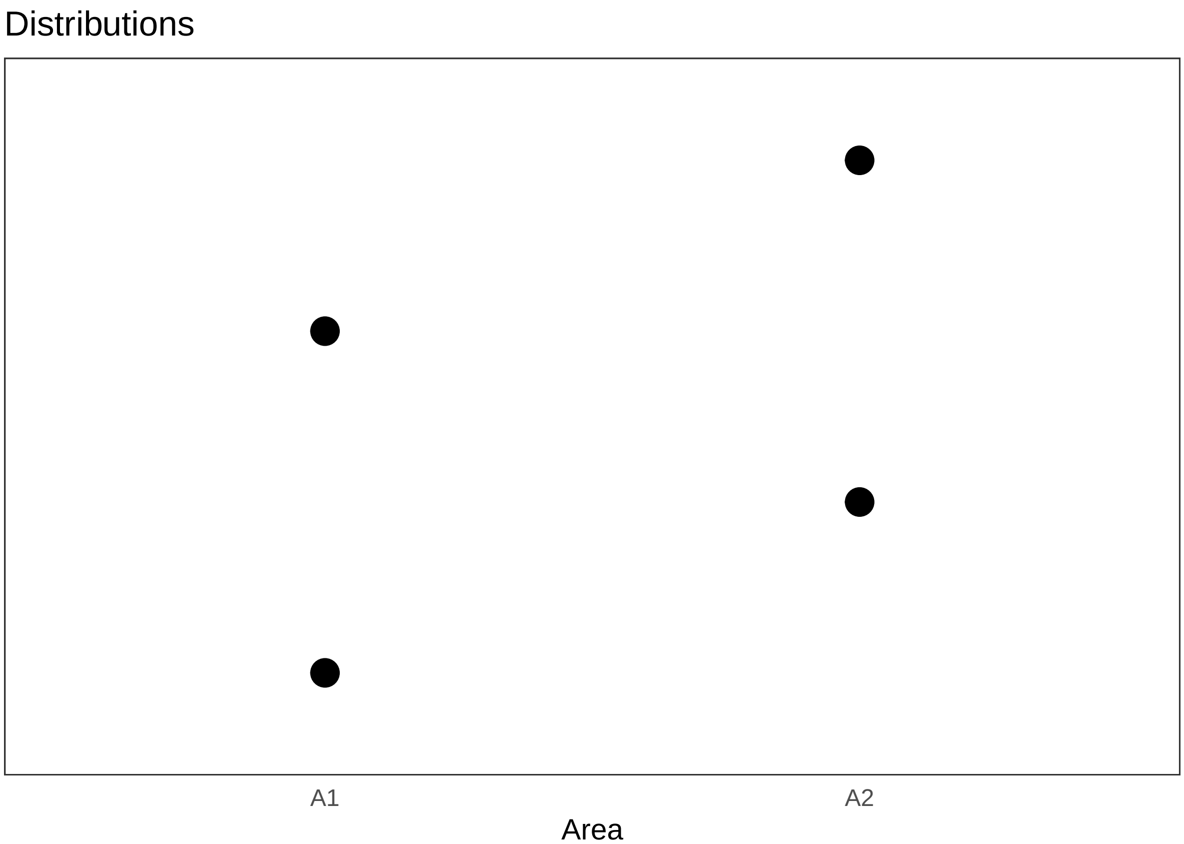
```r
plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle("Four terminals, all branches are == 1")


##
print(plotTree)
```



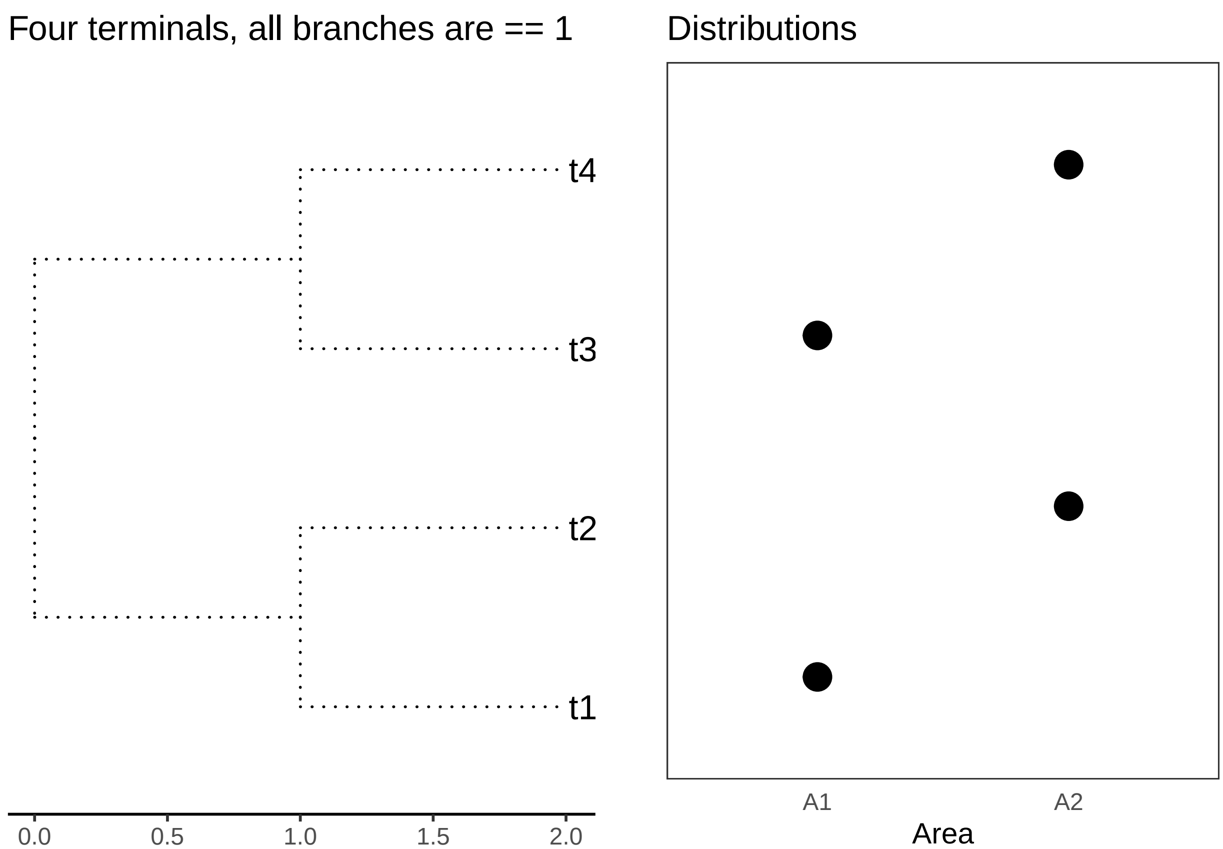

```r
## the distribution


plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150))     +
               geom_point()      +
               theme_bw()                                       +
               theme( axis.text.y = element_blank(),
                      panel.grid.major = element_blank(),  # Remove major gridlines
                      panel.grid.minor = element_blank(),  # Remove minor gridlines
                      axis.line = element_line(colour = NA_character_),  # Remove axis lines
                      axis.ticks = element_blank(),  # Remove axis ticks
                      legend.position = "none"
                      ) +
               labs(title = "Distributions",
                    y = "",
```

```
                        x = "Area")
```

```
##
print(plotDistrib)
```




```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```




```r
#We check whether names in both objects: initialTree and dist4taxa, are the same.


##all(colnames(dist4taxa) == initialTree$tip.label)

all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```r
#We report the branch length, and calculate the PD values.


initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```r
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)

initialPD
```

```
## [1] 4 4
```

```r
#As expected there is a tie between both areas.


#Now, we can see the impact of swapping the branch lengths; we will use the three
#available models to swap: "simpleswap","allswap","uniform", and we will swap the
#three classes of branches: "terminals","internals","all".
```

```r
##?swapBL


for( modelo in c("simpleswap","allswap","uniform") ){

        for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
            val <- swapBL(tree=initialTree,
                          distribution = dist4taxa,
                          model = modelo,
                         branch = rama
                         )
        printBlepd(val,tabular=TRUE,compact=TRUE)

         cat( "\n\n" )

     }

  }
```

```
## model to test simpleswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
```

```
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
```

```
val$bestModifiedArea
```

```
##   AreaSelected Freq
## 1         A1A2  100
```

```
t(val$bestModifiedArea)
```

```
##              [,1]
## AreaSelected "A1A2"
## Freq         "100"
```

```
data.frame(t(val$bestModifiedArea))
```

```
##              t.val.bestModifiedArea.
## AreaSelected                    A1A2
## Freq                             100
```

```
b <-  data.frame(t(val$bestModifiedArea))

colnames(b) <- b[1,]
                        b <- b[2,]
class(b)
```

```
## [1] "character"
```

```
str(b)
```

```
##  chr "100"
```

```
                        print(b)
```

```
## [1] "100"
```

```
##


#As expected, as all braches are EQUAL nothing happens
```

```r
#Now, let us see if the results could be related to the branch length (or not).



## we could use a single command (*evalBranch*)


##?evalBranch

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")
```

```
## Loading required package: maps

##
## Attaching package: 'phytools'

## The following object is masked from 'package:vegan':
##
##     scores
```

```r
## let's see the structure of the output

##str(AllTerminals)

## Now we can print a more friendly output


printEvalBranch(AllTerminals, compact=FALSE)
```

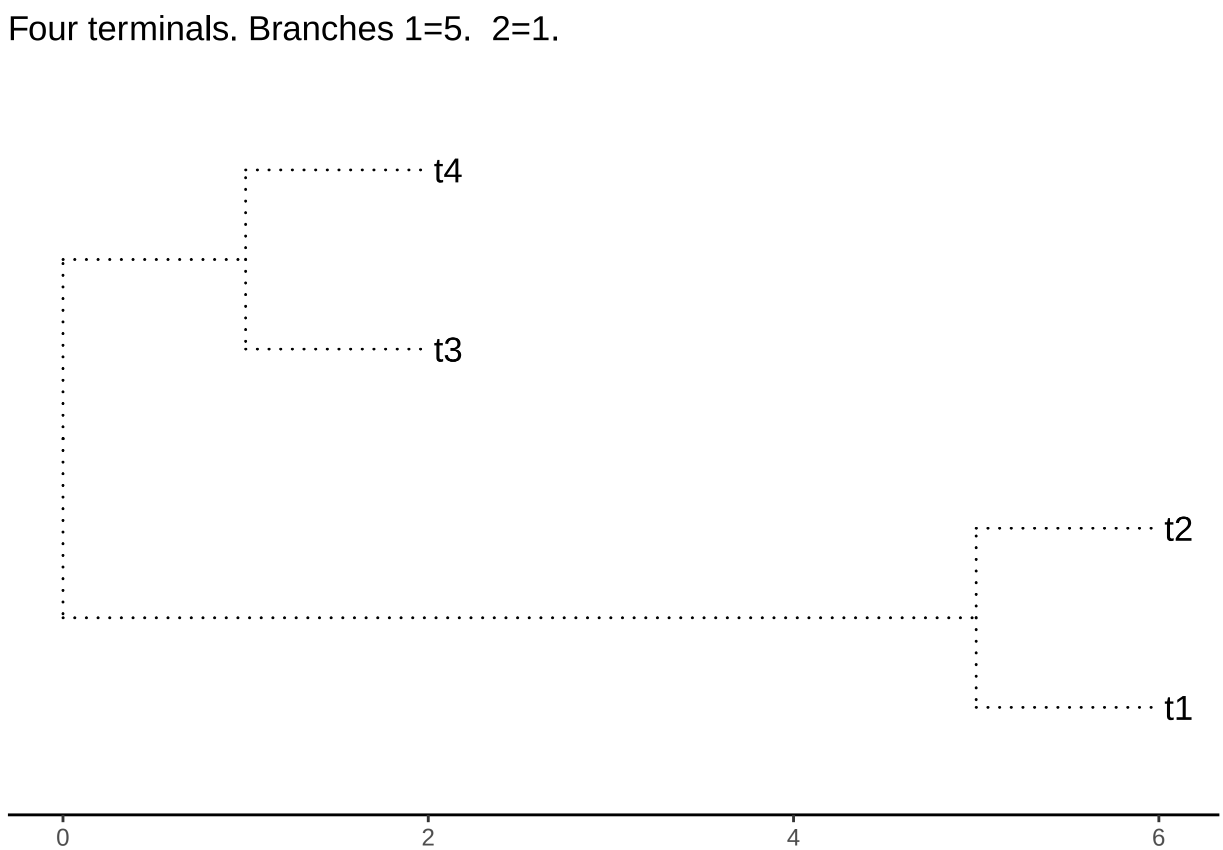
```
## data frame with 0 columns and 0 rows
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
#now with a longer branch


initialTree <- tree

initialTree$edge.length[1] <- 5

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
```

```r
            geom_tiplab(size=4.5, color="black") +
            theme_tree2() +
            ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
            ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```r
 for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )
       printBlepd(val,tabular=TRUE,compact=TRUE)

        cat( "\n\n" )

    }

 }
```

```
## model to test simpleswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##    2
## 1  "A1A2"
## 16 "74"
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##    2
## 1  "A1A2"
## 40 "38"
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 57 "43"
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

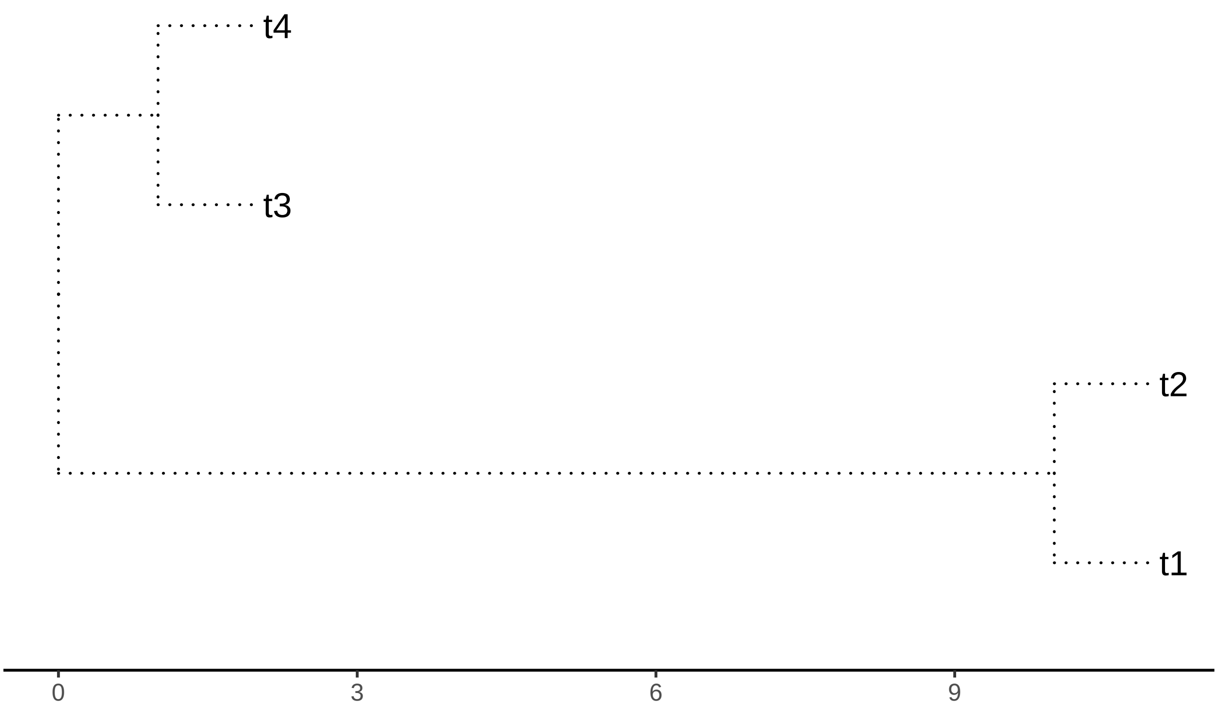
```
## data frame with 0 columns and 0 rows
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
#now with a longer branch


initialTree <- tree

initialTree$edge.length[1] <- 10


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=10. 2=1.




```
 for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                        branch = rama
                        )
       printBlepd(val,tabular=TRUE,compact=TRUE)

        cat( "\n\n" )

     }

  }
```

```
## model to test simpleswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##     NA
```

```
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##   2
## 1 "A1A2"
## 9 "75"
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##    2
## 1  "A1A2"
## 25 "39"
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 45 "55"
```

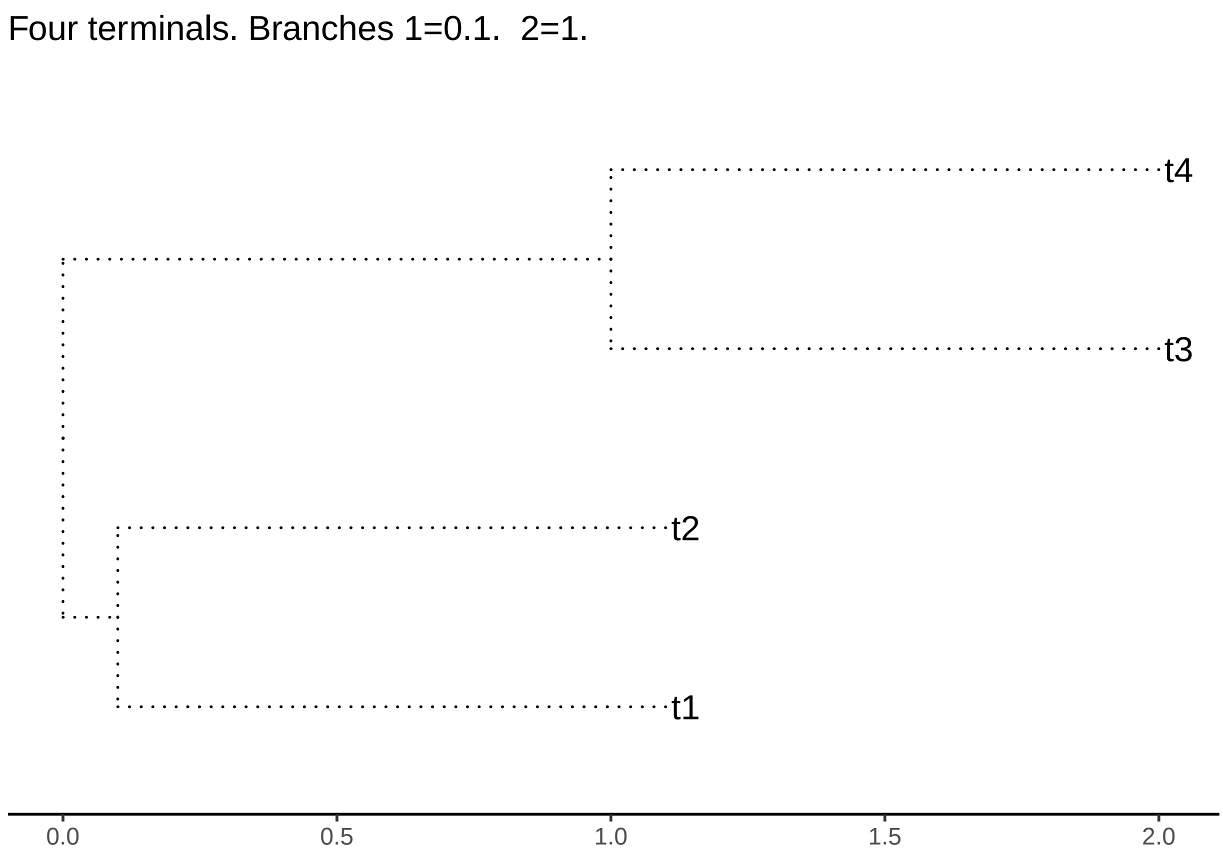
```r
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
#now with a longer branch



initialTree <- tree

initialTree$edge.length[1] <- 0.1


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```
 for( modelo in c("simpleswap","allswap","uniform") ){

        for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
            val <- swapBL(tree=initialTree,
                          distribution = dist4taxa,
                          model = modelo,
                         branch = rama
                         )
        printBlepd(val,tabular=TRUE,compact=TRUE)

         cat( "\n\n" )

    }

  }
```

```
## model to test simpleswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##     NA
```

```
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##    2
## 1  "A1A2"
## 12 "72"
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##    2
## 1  "A1A2"
## 36 "31"
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 46 "54"
```

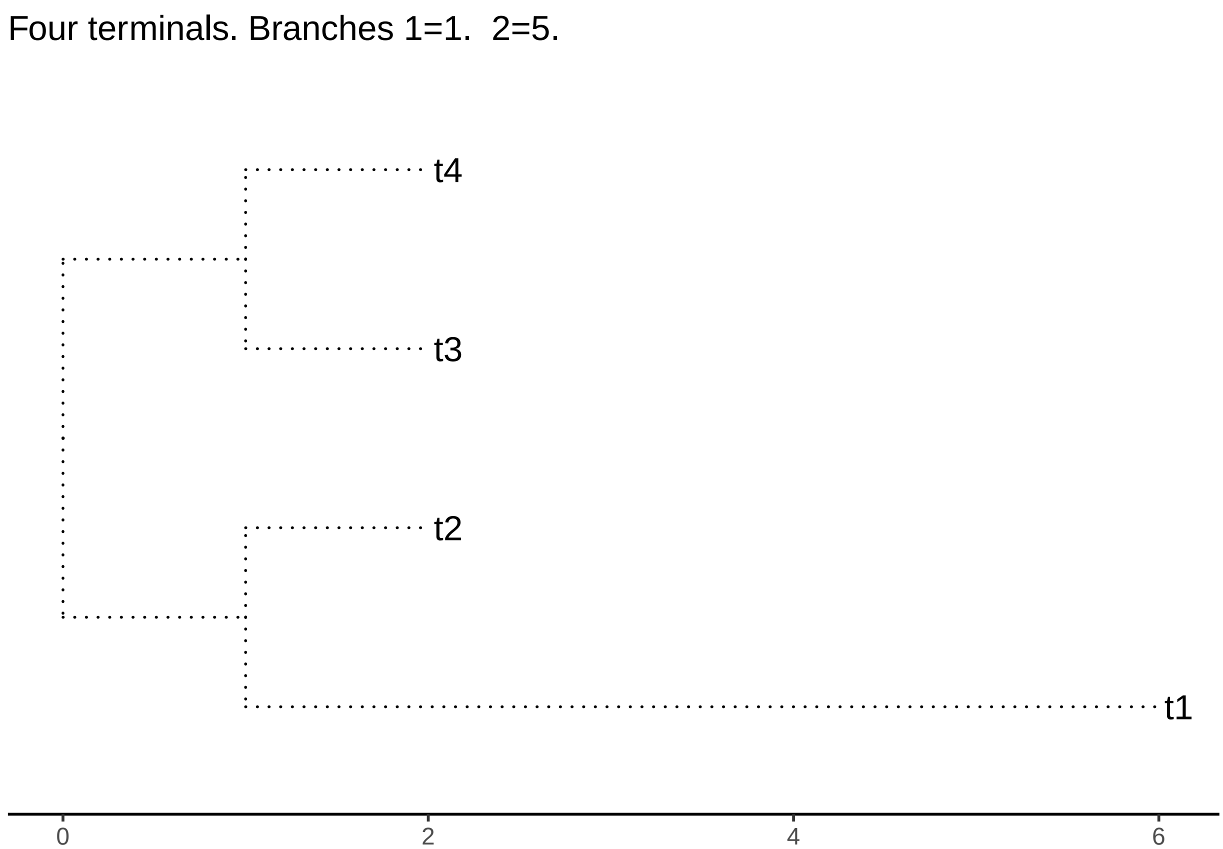
```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
#now with a longer branch in brach 2


#now with a longer branch


initialTree <- tree

initialTree$edge.length[2] <- 5

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```
for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                        branch = rama
                        )
       printBlepd(val,tabular=TRUE,compact=TRUE)

        cat( "\n\n" )

    }

 }
```

```
## model to test simpleswap reps 100
##    2
## 1  "A2"
## 68 "32"
##
##
## model to test simpleswap reps 100
##     NA
```

```
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##    2
## 1  "A1A2"
## 74 "15"
##
##
## model to test allswap reps 100
##    2
## 1  "A2"
## 54 "46"
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##    2
## 1  "A1A2"
## 37 "31"
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 51 "49"
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 51 "49"
```

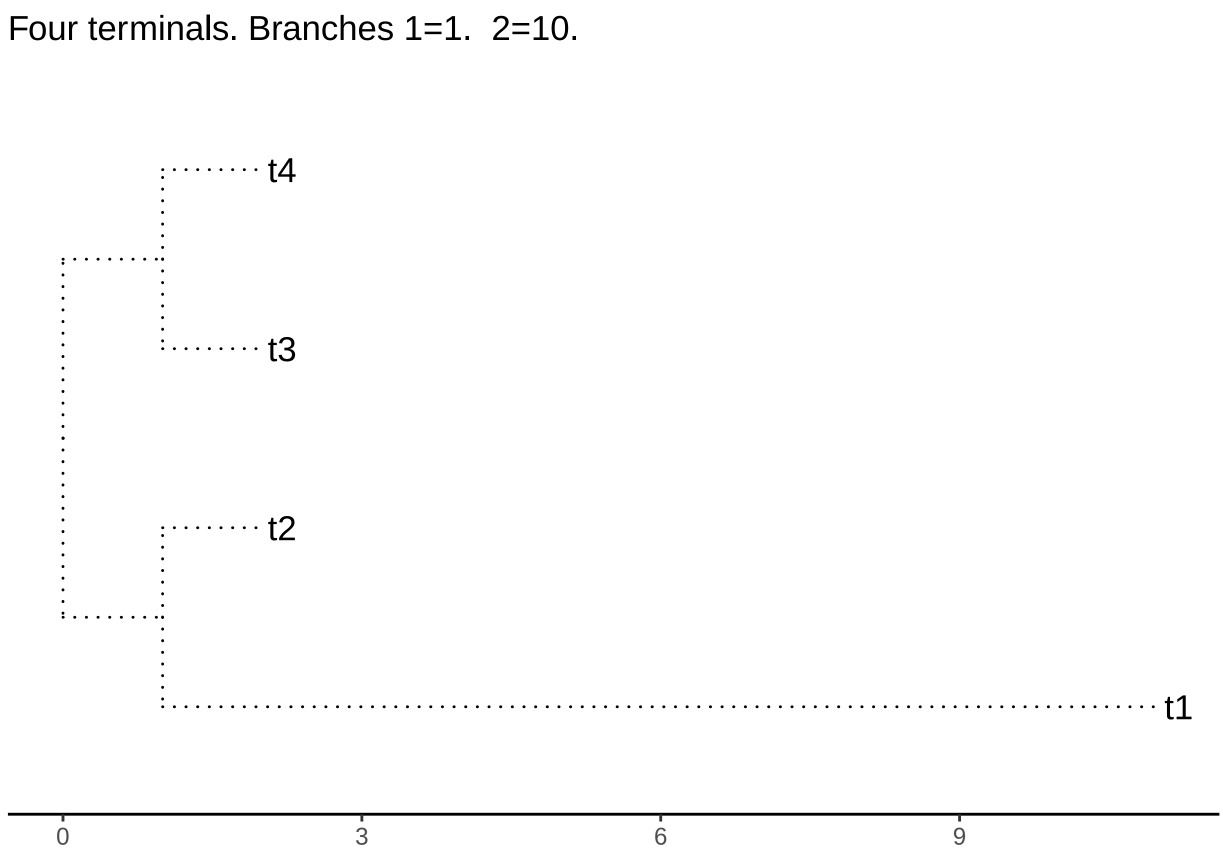
```r
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
#now with a longer branch


initialTree <- tree

initialTree$edge.length[2] <- 10


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```
 for( modelo in c("simpleswap","allswap","uniform") ){

        for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
            val <- swapBL(tree=initialTree,
                          distribution = dist4taxa,
                          model = modelo,
                         branch = rama
                         )
        printBlepd(val,tabular=TRUE,compact=TRUE)

         cat( "\n\n" )

     }

  }
```

```
## model to test simpleswap reps 100
##    2
## 1  "A2"
## 70 "30"
##
##
## model to test simpleswap reps 100
##     NA
```

```
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##    2
## 1  "A1A2"
## 80 "10"
##
##
## model to test allswap reps 100
##    2
## 1  "A2"
## 45 "55"
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##    2
## 1  "A1A2"
## 40 "31"
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 43 "57"
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 45 "55"
```

```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

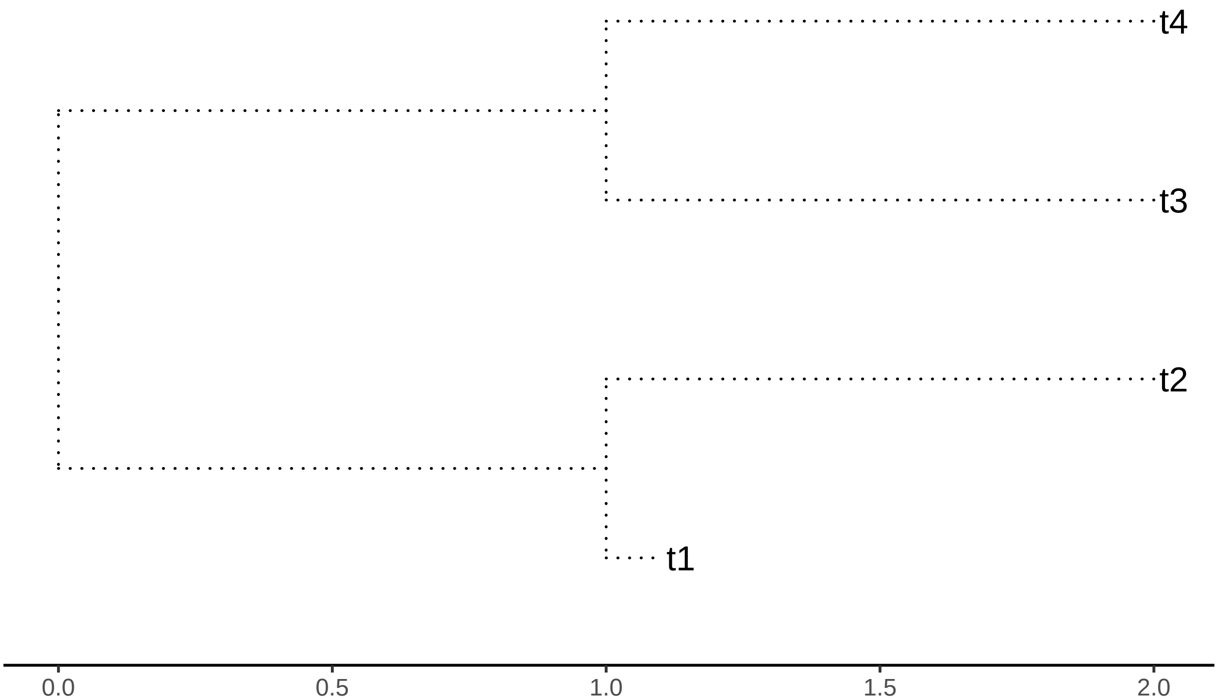
```
## data frame with 0 columns and 0 rows
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval="internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
#now with a longer branch



initialTree <- tree

initialTree$edge.length[2] <- 0.1


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=1. 2=0.1.




```
 for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                        branch = rama
                        )
       printBlepd(val,tabular=TRUE,compact=TRUE)

        cat( "\n\n" )

     }

  }
```

```
## model to test simpleswap reps 100
##    2
## 1  "A2"
## 23 "77"
##
##
## model to test simpleswap reps 100
##     NA
```

```
## 1   NA
## 100 NA
##
##
## model to test simpleswap reps 100
##    2
## 1  "A1A2"
## 16 "12"
##
##
## model to test allswap reps 100
##    2
## 1  "A2"
## 52 "48"
##
##
## model to test allswap reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test allswap reps 100
##    2
## 1  "A1A2"
## 33 "35"
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 45 "55"
##
##
## model to test uniform reps 100
##     NA
## 1   NA
## 100 NA
##
##
## model to test uniform reps 100
##    2
## 1  "A2"
## 53 "47"
```

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
## data frame with 0 columns and 0 rows
```